



Indira Gandhi Delhi Technical University for Women
(Established by Govt. of Delhi vide Act 09 of 2012)
Kashmere Gate, Delhi-110006

Bachelor of Technology
CSE
(Second Year)

Teaching Scheme & Syllabus

Academic Year: 2026-27

B.Tech. – Computer Science and Engineering					
Third Semester					
S No	Type of Course	Subject	L-T-P	Code	Credits
1.	Basic Sciences	Discrete Mathematics	3-0-0	BAS 201	3
2.	DCC	Software Engineering	3-0-2	BIT 205	4
3.	DCC	Design and Analysis of Algorithm	3-0-2	BCS 201	4
4.	Interdisciplinary	Fundamental of Analog & Digital Electronics	3-0-2	BEC 207	4
		Semiconductor Technology	3-0-2	BEC 208	
		Operations Management	3-1-0	BMA 215	
		Ergonomic Design	3-1-0	BMA 216	
		Measurement and Metrology	3-0-2	BMA 217	
		Solar Energy Technology	3-1-0	BMA 218	
		Industrial Waste Water Treatment	3-1-0	BMA 219	
		Numerical Methods	3-1-0	BAS 205	
		Optimization Techniques	3-1-0	BAS 206	
		Material Science and Engineering	3-1-0	BAS 207	
5.	AEC/VAC	Indian Knowledge System	2-0-0	AEC 201	2
6.	SEC	Introduction to Internet of Things	2-0-4	BCS 202	4
7.	Internship	Internship	0-0-2	BCS 250	1
		Total			22
Fourth Semester					
S No	Type of Course	Subject	L-T-P	Code	Credits
1.	DCC	Database Management Systems	3-0-2	BAI 201	4
2.	DCC	Computer Organization and Architecture	3-0-2	BCS 203	4
3.	DCC	Operating Systems	3-0-2	BIT 204	4
4.	DEC	Data Communication and Computer Networks	3-0-2	BCS 204	4
		Information Theory and Coding	3-0-2	BEC 212	
		Statistical Modeling	3-0-2	BIT 206	
		Data Mining and Data Warehouse	3-0-2	BAI 204	
		Numerical Methods	3-1-0	BAS 205	
		Optimization Techniques	3-1-0	BAS 206	
		Material Science and Engineering	3-1-0	BAS 207	
5.	AEC/VAC	Human Values and Professional Ethics	2-0-0	AEC 202	2
6.	SEC	Advanced IoT and Real World Applications	2-0-4	BCS 205	4
		Total			22

DISCRETE MATHEMATICS	
Course Code: BAS 201 Contact Hours: L-3 T-0 P-0 Course Category: Basic Sciences	Credits: 3 Semesters: 3

Introduction: The discrete structures subject introduces Propositional logic, Sets, Relations, and Functions, Algebraic structures, Graphs and Trees required for building mathematical foundation of computer science.

Course Objective:

1. To introduce and understand the fundamental notions in discrete mathematics.
2. To understand basic concept of an algorithm and its application in combinatorial mathematics.
3. To introduce the basic properties of graphs and trees and model simple applications.

Prerequisite: Basic concepts of set theory.

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Understand a logic sentence in terms of predicates, quantifiers, and logical connectives and its validation.

CO2: Able to apply logical notations to define and reason about fundamental mathematical concepts such as sets relations, functions and combinatorics.

CO3: Able to apply logical notations to define and reason about fundamental mathematical concepts of abstract algebra.

CO4: Apply algorithms and use of graphs and trees as tools to analyse and simplify problems.

Pedagogy:

The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations, and quizzes. Students would be encouraged to develop an understanding of the subject. The use of ICT and web-based sources will be adopted.

Contents

UNIT-I	10 Hours
Propositional logic: Syntax, semantics, valid, satisfiable and unsatisfiable formulas, propositions, negation disjunction and conjunction, implication and equivalence, truth tables, predicates quantifiers, natural deduction, Rules of Inference. Methods of proofs: Forward proof, proof by contradiction, contra positive proofs, proof of necessity and sufficiency.	
UNIT-II	10 Hours
Sets, relations and functions: Operations on sets, relations, binary relations, partial ordering relations, equivalence relations and partitions, Partial orderings, Posets, Linear and well-ordered sets, principles of mathematical induction, Finite and infinite sets, countable and uncountable sets, power set theorem. Functions, mappings, injection and surjection, composition of functions, inverse functions, special functions, pigeonhole principle, recursive function theory.	
UNIT-III	10 Hours
Algebraic structures and Morphisms: Algebraic structures with one binary operation - semigroups, monoids and groups, subgroups and their properties, congruence relation. Free and cyclic monoids and groups, permutation groups, substructures, normal subgroups. Algebraic structures with two binary operations - rings, integral domains and fields.	
UNIT-IV	10 Hours
Graphs and trees: Terminology, Graphs and basic properties - degree, path, cycle, subgraphs, isomorphism, Eulerian and Hamiltonian walks, Graph coloring, planar graphs, directed graphs, trees terminology, tree traversals, spanning trees.	
Text Books	
1.	Kenneth H Rosen, “Discrete Mathematics and Its Applications”, McGraw Hill, 8 th Edition.
2.	C L Liu, D P Mohapatra, “Elements of Discrete Mathematics: A Computer Oriented Approach”, McGraw Hill, 4 th Edition.
3.	Bernard Kolman, Robert Busby, Sharon C. Ross, “Discrete Mathematical Structures”, Prentice Hall of India, 6 th Edition.
Reference Books	
1.	Narsingh Deo, “Graph Theory with Applications to Engineering and Computer Science” (ISBN: 9789390669509).
2.	Norman L Biggs, “Discrete Mathematics”, Oxford University Press, 2 nd Edition.
3.	J P Trembly & P Manohar, “Discrete Mathematical Structures with Applications to Computer Science”, McGraw Hill, 1987.

Software Engineering	
Course Code: BIT 205 Contact Hours: L-3 T-0 P-2 Course Category: DEC	Credits: 4 Semesters: 4

Introduction: The course introduces students to the different software development lifecycle (SDLC) phases used in developing, delivering, and maintaining software products. Students will also acquire basic software development skills and understand common terminology used in the software engineering profession. The aim of the course is to provide an understanding of the working knowledge of the techniques for estimation, design, testing and quality management of software development projects.

Course Objectives:

1. To introduce the concepts of software engineering, software processes and its models.
2. To understand the software requirements analysis, transform the requirements using DFD, create software requirement specification document and validation of the software requirements.
3. To understand fundamental of software design, software quality and software maintenance.
4. To understand the project planning process, size and cost estimation techniques for development of software.

Course Outcomes: At the end of the course, the students will be able to:

CO1: Understand the concepts of Software engineering, Software process and its models.

CO2: Evaluate the Software Requirements, interpret and structure the requirements in Software Requirement Document.

CO3: Apply appropriate software architectures and patterns to carry out high level design of a system and be able to critically compare alternative choices, evaluate the quality and maintenance of the software through software testing.

CO4: Create the software project plan for size and cost estimation including risk analysis

Pedagogy: Teaching pedagogy for Software Engineering will incorporate a mix of theory, practical application, case studies, and hands-on projects to enhance student learning and engagement. The student will learn basics of UML Modeling and Automation of Software Testing in the Lab using open source software tools.

UNIT-I: Introduction		11Hours
Introduction to Software Engineering: Definition and Importance of Software Engineering, Software Crisis and Myths Software Process Framework, Software Development Life Cycle (SDLC) Models : Waterfall Model, Incremental Model, Prototype Model Spiral Model, Agile Model.		
UNIT-II: Modeling		11 Hours
Software Requirement Engineering: Understanding Software Requirements, Requirement Engineering Process, Requirements Analysis, Requirements Modeling : DFD, Use Case Modeling, Software Requirement Specification (SRS), Software Design & Architecture: Software Design Principles, Coupling and Cohesion, Design Pattern, Refactoring Architectural Styles & Patterns		
UNIT-III: Quality Management		11 Hours
Software Testing: Software Testing Fundamentals, Testing Techniques: White Box, Black Box, and Grey Box Testing, Unit Testing, Integration Testing, System Testing, Regression Testing, Automated Testing Tools : Selenium, JUnit, Web Application Testing, Security Testing		
UNIT-IV: Managing Software Projects		11 Hours
Software Project Management: Project Planning, FPA Model, Software Cost Estimation (COCOMO Model), Risk Management, Configuration Management, Software Maintenance: Software Maintenance Types,Software Maintenance Models,Reverse Engineering & Re-engineering, Change Management Process		
Text Books		
1	Roger S. Pressman: Software Engineering: A Practitioner’s Approach, 9th Edition 2023, McGraw Hill	
2	KK Aggarwal, Yogesh Singh :Software Engineering, 4thEdition, 2022, New Age International Publishers	
Reference Books		
1	Ian Sommerville: Software Engineering, 10th Edition 2017, Pearson Education.	
2	Paul C. Jorgenson, Byron DeVries: SoftwareTesting- A Craftsman Approach, 5th Edition, 2021, CRC Press.	

DATABASE MANAGEMENT SYSTEMS	
Course Code: BAI 201 Contact Hours: L-3 T-0 P-2 Course Category: DCC	Credits: 4 Semesters: 3

Introduction:

Database Management System (DBMS) is used for creating and managing the databases. The main aim of a DBMS is to supply a way to store-up and retrieve the desired database information as per the application requirement, which is both convenient and efficient.

Course Objective:

- To introduce the concepts of Database Management Systems and designing the relational databases by applying normalization techniques to normalize the database
- Strong practice in SQL programming through a variety of database problems.
- Understand the needs of database processing and learn techniques for controlling the consequences of concurrent data access and recovery from failures.
- To write PL/SQL programs using Cursors, Exception, Procedures, Functions and Triggers

Prerequisite:

Concepts of basic Mathematics and Programming

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Understand fundamentals of Database Management Systems and design Database models and learn database Languages to write Queries to extract information from Databases.

CO2: Construct and interpret SQL commands to create tables and indexes, insert/update/delete data, and query data in a relational DBMS.

CO3: Identify Database Anomalies and improve the design of Database Management System

CO4: Understand Transaction Management and Concurrency control

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.

Contents

UNIT-I		10 Hours
Overview of Concepts and Conceptual Database Design, Characteristics of the Database, DBMS Architecture, File System vs Database System, Database Administrator and Database Users, Data Models, Schemes and Instances, Data Independence, Database Languages and Interfaces, Data Models		
UNIT-II		11 Hours
Data Modeling: Entity-Relationship Model, Strong and Weak Entity Sets, Generalization, Specialization, and Aggregation, Relational Model, Languages and Systems: Relational Model Concepts, Relational Model Constraints, Translating your ER Model into Relational Model, Relational Algebra. SQL: A Relational Database Language, Data Definition/Manipulation/Control in SQL, Relational Model Constraints, integrity Constraints, Key Domain Constraints, referential Integrity, Foreign key. Indexes in SQL, View and Queries in SQL, Practicing SQL commands		
UNIT-III		11 Hours
Relational Database Design: Functional Dependencies & Normalization for Relational Databases, Functional Dependencies, Normal Forms (1NF, 2NF, 3NF, BCNF, 4NF, 5NF), Lossless Join and Dependency Preserving Decomposition, Multivalued Dependency, Join Dependency. Transaction Management: Transaction Concept and State, ACID Properties, Concurrency Control: Lock-Based Protocols, Timestamp-based Protocols, Recovery from Transaction Failures, Log based Recovery, Checkpoints, Deadlock Handling		
UNIT-IV		10 Hours
Database Programming: Exceptions, Cursors, Procedures, Functions, Triggers NoSQL Databases: Introduction, properties of NoSQL Databases, Types of NoSQL databases, Introduction to MongoDB.		
Text Books		
1.	Elmasri Ramez and Navathe Shamkant, Fundamentals of Database System, Pearson, 7 th /Latest Edition (2017).	
2.	Abraham Silberschatz, Henry F. Korth, S. Sudarshan, Database System Concepts, McGraw Hill, 7 th /Latest Edition (2021)	
Reference Books		
1.	Date C.J., "An Introduction to Database systems", Addison-Wesley Longman, Inc., 8th/ Latest Edition (2006)	
2.	Raghu Ramkrishnan and Johannes Gehrke, Database Management Systems, McGraw-Hill, 3 rd /Latest Edition (2003)	

DATA MINING AND DATA WAREHOUSE	
Course Code: BAI 204 Contact Hours: L-3 T-0 P-2 Course Category: DEC	Credits: 4 Semesters: 4

Introduction: This subject introduces the concepts of data warehousing, a system for storing and managing large amounts of data from various sources, and data mining, a process of extracting meaningful patterns and insights from that data through advanced analytical techniques. The course will cover key aspects like data warehouse architecture, data preprocessing, various data mining algorithms (classification, clustering, association rule mining), and evaluation metrics, enabling students to understand how to build and utilize data warehouses for extracting valuable business insights.

Course Objective:

1. Be familiar with mathematical foundations of data mining tools.
2. Understand and implement classical models and algorithms in data warehouses and data mining.
3. Characterize the kinds of patterns that can be discovered by association rule mining, classification and clustering.
4. Master data mining techniques in various applications like social, scientific and environmental context.
5. Develop skill in selecting the appropriate data mining algorithm for solving practical problems.

Prerequisite: NIL

Course Outcomes:

After successful completion of the course, the student will be able to:

- CO1:** Understand the functionality of the various data mining and data warehousing components.
- CO2:** Analyze the strengths and limitations of various data mining and data warehousing models.
- CO3:** Explain the analyzing techniques of various data.
- CO4:** Describe different methodologies used in data mining and data warehousing and compare with various technologies.
- CO5:** Create real world projects using concepts of data warehousing and data mining.

Pedagogy: Chalk and talk method, PowerPoint Presentation, Demonstration, Problem based learning.

Contents

UNIT-I		11 Hours
The Compelling Need for data warehousing. Data warehouse - The building Blocks Defining the business requirements, Requirements definition (scope and content), Principles of dimensional modelling, Dimensional Modelling. OLAP in the Data Warehouse: Demand for Online analytical processing, need for multidimensional analysis, OLAP characteristics, features and functions, dimensional analysis, hyper cubes, Drill-down and roll-up, slice-and-dice, rotation. OLAP models: MOLAP model, ROLAP model, HOLAP model, ROLAP versus MOLAP OLAP implementation considerations.		
UNIT-II		11 Hours
Data Mining Basics: Data Mining Definition. The knowledge discovery process, OLAP versus data mining, data mining and the data warehouse, Process of data mining. Data Mining Applications. Benefits of data mining. Associations and Correlations: Association rule mining, Apriori algorithm, improving efficiency, kinds of association rules, multilevel, multi-dimensional.		
UNIT-III		11 Hours
Classification and Regression: types of classification algorithm, Bayesian Classification, rule based classification, KNN, Classification by Decision Tree Introduction, Classification by Back propagation, Support Vector Machines, Lazy Learners, Prediction Techniques, Regression Models.Cluster analysis: Overview of grid based, model based. density based, partitioning based, hierarchical based clustering methods.		
UNIT-IV		11 Hours
Major Data Mining Techniques: Cluster detection, K-means algorithm, link analysis, neural networks, genetic algorithms, fuzzy logic, Agglomerative Hierarchical Clustering, DBSCAN, Cluster Evaluation, Density-Based Clustering, Graph-Based Clustering, Scalable Clustering Algorithms.Web Mining, Sentiment Analysis, Opinion Mining		
Text Books		
1	M.H. Dunham, "Data Mining Introductory and Advanced Topics", Pearson Education,2008	
2	Jawei Han, Micheline Kamber, and Jian Pei, Data Mining: Concepts and Techniques, 3rd Edition, Morgan Kaufmann, 2011.	
Reference Books		
1	Pieter Adriaans, Dolf Zantinge, "Data Mining", Pearson Education Asia, 2011	
2	Ralph Kimball. "The Data Warehouse Lifecycle toolkit", John Wiley, 2 edition, 2007	
3	M Berry and G. Linoff, "Mastering Data Mining", John Wiley, 3 edition, 2011	
4	Paul Raj Poonia, "Fundamentals of Data Warehousing", John Wiley & Sons, 2007	

OPERATING SYSTEMS	
Course Code: BIT 204 Contact Hours: L-3 T-0 P-2 Course Category: DCC	Credits: 4 Semesters: 4

Introduction:

This course will aim at introducing classical internal algorithms and structures of modern operating systems including CPU scheduling, memory management, and device management. Topics including file systems, virtual memory, disk request scheduling, concurrent processes, deadlocks, security, and integrity will be covered.

Course Objective:

1. To learn the fundamentals of Operating Systems & the mechanisms of OS to handle processes and their communication.
2. To learn the mechanisms involved in memory management.
3. To gain knowledge on OS architecture, mutual exclusion algorithms, deadlock detection algorithms etc.

Prerequisite: Basic programming knowledge in C or C++.

Course Outcomes:

After successful completion of the course, the student will be able to:

- CO1:** To understand various types of OS, basic concepts, various functions of different OS, process management & CPU scheduling
- CO2:** To compare and contrast various memory management schemes like paging, segmentation and to apply different deadlock handling algorithms
- CO3:** To implement different disk scheduling algorithms, to apply and use various process synchronization techniques and device management strategies.
- CO4:** To analyse management of I/O and different file handling & directory implementation schemes OS.

Pedagogy: The class will be taught using theory and tutorial-based methods which include board teaching and presentations/slides, discussions, brainstorming, case studies etc. Along with classroom teaching, students will also be given assignments regarding the topics covered.

Contents

UNIT-I	11 Hours
Introduction: Introduction to Operating System, Types of O.S: Simple Batch, Multi-programmed Batched, Time-Sharing, Personal-computer, Parallel, Distributed, Real-Time, Mobile Operating-System Structures: Layered Architecture, System Calls, System Programs, System Structure, Virtual Machine Processes: Process Concept, Process Scheduling, Operations on Processes, Cooperating Processes, Inter-process Communication, Threads, Multithreaded Programming. CPU Scheduling: Basic Concepts, Scheduling Criteria, Scheduling Algorithms, Multiple-Processor Scheduling, Real-Time Scheduling	
UNIT-II	11 Hours
Process Synchronization: Background, Critical-Section Problem, Synchronization Hardware, Semaphores, Classical Problems of Synchronization, Critical Regions, Monitors. Memory Management: Background, Logical versus Physical Address space, Swapping, Contiguous allocation, Fragmentation, Paging, Segmentation, Segmentation with Paging. Virtual Memory: Demand Paging, Page Replacement, Page-replacement Algorithms, Performance of Demand Paging, Allocation of Frames, thrashing. Deadlocks: System Model, Deadlock Characterization, Methods for Handling Deadlocks, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Recovery from Deadlock	
UNIT-III	11 Hours
Device Management: Techniques for Device Management, Dedicated Devices, Shared Devices, Virtual Devices Secondary-Storage Structure: Disk Structure, Disk Scheduling, Disk Management, Swap-Space Management, Disk Reliability, Stable-Storage Implementation	
UNIT-IV	11 Hours
Information Management: Introduction, Simple File System, General Model of a File System, Symbolic File System, Basic File System, Access Control Verification, Logical File System, Physical File System File-System Interface: File Concept, Access Methods, Directory Structure, Protection, and Consistency Semantics. File-System Implementation: File-System Structure, Allocation Methods, Free-Space Management, Directory Implementation, Efficiency and Performance, Recovery.	
Text Books	
1	Silberschatz and Galvin, “Operating System Concepts”, John Wiley, 9th Ed., 2016/Latest Edition.
2	Madnick E. and Donovan J., “Operating Systems”, McGraw Hill, 2017/Latest Edition.
3	Tannenbaum, “Operating Systems”, PHI, 5th Ed/Latest Edition.
Reference Books	
1	Milnekovic, “Operating System Concepts”, McGraw Hill, 1992
2	Harvey M Dietel, “ An Introduction to Operating System”, Pearson Education, 2004

Statistical Modelling	
Course Code: : BIT 206 Contact Hours: L-3 T-0 P-2 Course Category: DEC	Credits:4 Semesters: 4

Introduction: Statistical Modelling

Study of this course provides the learners a clear understanding of fundamental concepts in probability and statistics. This course covers the modern theory of probability and statistics, important models of sampling, techniques of hypothesis testing and correlation & regression. The course helps the learners to find varied applications in engineering and science like disease modelling, climate prediction and computer networks.

Course Objectives

1. Understand probability distributions, random variables, and their statistical properties.
2. Learn sampling techniques, estimation methods, and sample size determination.
3. Conduct hypothesis testing, ANOVA, and statistical test procedures.
4. Analyse relationships using correlation, regression, and model adequacy assessment.

Course Outcome

At the end of the course, the students will be able to:

- CO1:** Explain the concept, properties and important models of discrete random variables and use them to analyse suitable random phenomena. Summarize the properties and relevant models of continuous random variables and use them to analyse suitable random phenomena.
- CO2:** Make use of concepts of sampling and theory of estimation to solve application-level problems.
- CO3:** Organize the basic concepts in hypothesis testing and develop decision procedures for the most frequently encountered testing problems.
- CO4:** Build statistical methods like correlation and regression analysis to interpret experimental data.

Pedagogy: The teaching pedagogy combines theoretical concepts with practical applications through lectures, problem-solving exercises, real-world case studies, and hands-on data analysis. Interactive discussions, assignments, and software-based demonstrations enhance understanding, while quizzes and assessments ensure continuous learning.

Contents

UNIT-I		10 Hours
Probability Distributions and Random Variables: Definition of random variables, Discrete vs. continuous random variables, Probability distributions of random variables, Definition of expectation, mean and variance for discrete and continuous distributions, Discrete Probability Distributions: Binomial distribution, Poisson distribution, Poisson, Continuous Probability Distributions: Uniform distribution, Exponential distribution, Normal distribution, Continuous bivariate distributions, Marginal distributions, Independent continuous random variables.		
UNIT-II		10 Hours
Sampling Techniques: Need for Sampling, Some Fundamental Definitions, Important Sampling Distributions, Sampling Theory, Sandler's A-test, Concept of Standard Error, Estimation, Estimating the Population Mean(μ), Estimating Population Proportion, Sample Size and its Determination, Determination of Sample Size through the Approach Based on Precision Rate and Confidence Level, Determination of Sample Size through the Approach Based on Bayesian Statistics.		
UNIT-III		10 Hours
Testing of Hypothesis: Hypothesis and Test Procedures, Tests about a population mean, Tests concerning a population proportion, p-values, Single factor ANOVA, F-test, Multiple comparisons in ANOVA, two factor ANOVA.		
UNIT-IV		10 Hours
Correlation and Regression Analysis: Simple Linear Regression Model, estimating model parameters, Correlation, Non-Linear and multiple regression, Assessing Model Adequacy, Regression with transformed values, Polynomial Regression, Multiple Regression Analysis		
Text Books		
1	Jay L. Devore, Probability and Statistics for Engineering and the Sciences, 8th edition, Cengage, 2012 2.	
2	Research Methodology: Methods and Techniques: C.R. Kothari, New Age International Publishers	
Reference Books		
1	HosseinPishro-Nik, Introduction to Probability, Statistics and Random Processes, Kappa Research, 2014	
2	Sheldon M. Ross, Introduction to probability and statistics for engineers and scientists, 4th edition, Elsevier, 2009.	
3	T. VeeraRajan, Probability, Statistics and Random processes, Tata McGraw-Hill,2008	

OPERATION MANAGEMENT	
Course Code: BMA 215 Contact Hours: L-3 T-1 P-0 Course Category: Interdisciplinary	Credits: 4 Semesters: 3

Introduction:

Operations Management (OM) is an interdisciplinary field that plays a crucial role in optimizing the efficiency and effectiveness of business operations. This course integrates principles from mechanical, industrial, and supply chain domains. The subject bridges the gap between technology and management, focusing on data-driven decision-making, process optimization, supply chain strategies.

Prerequisite: Elements of Mechanical Engineering

Course Outcomes:

After successful completion of the course, the student will be able to:

- CO1: **Understand** the principles of productivity and organizational structures, capacity planning, plant location, and layout principles
- CO2: **Analyze** different forecasting techniques, and their relevance in operations management, method study and work study techniques.
- CO3: **Analyze** inventory management techniques, including EOQ models, and apply them to manage stock levels effectively.
- CO4: **Evaluate** the key components of production planning and supply chain management to improve operational performance.

Pedagogy: Classroom teaching is supported by White board, black board, chalks, markers, projector and screen. The hand written notes, PowerPoint slides and assignments will be provided to the students. The students can also raise their issues related to the course in the class.

Contents

UNIT-I	10 Hours
Introduction –Introduction to productivity, Multi Factor productivity, Principles of management, Organization structure. Capacity Planning, Plant Location and Plant Layout – Introduction, need for selecting a suitable location, Location Factors, Quantitative Method, Principles of Plant layout, Types of Layout – Product, Process, Fixes Position, Cellular Layout.	
UNIT-II	10 Hours
Demand Forecasting-Need for demand forecasting, Techniques of forecasting, Time series analysis, Least Square Method, Moving Average, Exponential Method and Qualitative Techniques. Method Study- Introduction, Objectives Steps, Micro motion Study, Cycle graph and chrono cycle graph, Therbligs and SIMO charts. Work Study – Objectives, Different Techniques, Standard Time, Allowances, Time study Numerical, Performance Rating, Work sampling. Process and Product Life Cycle,	
UNIT-III	10 Hours
Inventory Management - Introduction, Reasons for Holding Inventories, Relevant Costs of Inventories, EOQ models, Quantity Discount Models, Safety Stock, Inventory control system, Selective Control of Inventory ABC analysis, VED analysis. Production Cost Concepts – Introduction, Cost of Production, Classification and analysis of cost, break even analysis, Make and Buy.	
UNIT-IV	10 Hours
Material Requirement Planning – Introduction, MRP objectives, Functions served by MRP Production Planning and Control, Supply chain and Logistics Management, Production Scheduling.	
Text Books	
1.	Martinich, J.S., Production and Operations Management: An Applied Modern Approach”, John Wiley and Sons, New Delhi, 2008.
2.	Richard B. Chase, Nicholas J.A., Jacobs, F.R., “Production and Operation Management”, Tata McGraw Hill, New Delhi, 1998
3.	Ravi Shankar, “Industrial Engineering and Management”, Galgotia Publications Pvt Ltd., 2000.
Reference Books	
1.	Paneerselvam, R., “Production and Operations Management”, Prentice Hall India, 2012.
2.	Khanna, O.P., “Industrial Engineering and Management”, Dhanpat Rai & Sons, 1985.

ERGONOMIC DESIGN	
Course Code: BMA 216 Contact Hours: L-3 T-1 P-0 Course Category: Interdisciplinary	Credits: 4 Semesters: 3

Introduction: An introduction to ergonomics affording students the necessary knowledge essential for the psychological and anthropometrical development leading to good design. Emphasis is placed on health and safety.

Prerequisite: Basic knowledge of statistics; means, standard deviations, and percentiles

Pre-Requisites: BMA-130 Engineering Graphics

Course Outcomes: After successful completion of the course, the student will be able to:

CO1: Knowledge of the sciences of human factors and workplace ergonomics.

CO2: Evaluation of the musculoskeletal disorders.

CO3: Knowledge about the cognitive aspects of user- system interaction

CO4: Analyze the environmental factors influencing human performance

Pedagogy: Classroom teaching is supported by White board, black board, chalks, markers, projector and screen. The hand written notes, PowerPoint slides and assignments will be provided to the students. The students can also raise their issues related to the course in the class.

Contents

UNIT-I	10 Hours
Definition, History, and Scope of Human Factors/Ergonomics; Principles of fitting design configurations to the users; Man-machine-environment interaction system: A design perspective; Human comfort, and adaptability.	
UNIT-II	10 Hours
Physical aspects of human factors: anthropometrics, human body structure and function, posture, joint movement, and biomechanics. Occupational stress and Musculoskeletal disorders; Safety and health issues.	
UNIT- III	10 Hours
Cognitive aspects of user-system interaction: perception, information processing, user behavior, error, and risk perception.	
UNIT-IV	10 Hours
Visual display in different planes- static shape, size, font type, and dynamic characters of display; Environmental factors influencing human performance.	
Text Books	
1.	Kroemer, K.H.E., “Fitting the Human: Introduction to Ergonomics”, 7th Edition, CRC Press, 2017.
2.	Pheasant, S. & Haslegrave, C., “Bodyspace: Anthropometry, Ergonomics, and the Design of Work”, 3rd Edition, CRC Press, 2005.
3.	K. H. E. Kroemer, H. B. Kroemer, Katrin E. Kroemer-Elbert, “Ergonomics: How to Design for Ease and Efficiency”, 2nd Edition, Prentice Hall, 2000
4.	Andris Freivalds, Benjamin Niebel “Methods, Standards and Work Design”, 13th Edition, McGraw Hill, 2013.
Reference Books	
1.	Marcelo M. Soares, Francisco Rebelo “Ergonomics in Design Methods and Techniques”, 1 st Edition, CRC Press, 2016.
2.	Jordan, P., “An Introduction to Usability”, 1st Edition, Taylor & Francis, 1998.

MEASUREMENT AND METROLOGY	
Course Code: BMA 217 Contact Hours: L-3 T-0 P-2 Course Category: Interdisciplinary	Credits: 4 Semesters: 3

Introduction:

This is a basic introductory course on measurement and metrology to be used in industry. For a product to be successful, it must be manufactured as per specifications and requirements. This course enlightens the students about different errors, calibration, sensors, accuracy of measurements thus to help in standardising the methods.

Course Objective:

- To enlighten the students on various measurement processes and their importance.
- To explain different types and sources of errors during measuring of different quantities.
- To determine the best way to do measurement and develop standardization of measuring methods.

Prerequisite: NIL

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Analyze measurements statistically and understand different sources of error

CO2: Understand measurement of different quantities such as temperature, pressure, flow, etc.

CO3: Apply concept of limits, fits and tolerances to calculate the dimensions of a product

CO4: Evaluate surface roughness, texture and roundness of a part

Pedagogy: The teaching-learning process comprises of lectures, tutorials, assignments, experiments, discussion, projects/ presentations and quizzes. Interactive classes help students to correlate theoretical concepts with practical examples for better understanding. Use of ICT and web-based resources will be utilized.

Contents

UNIT-I		10 Hours
Introduction: Introduction to measurement and measuring instruments, units of measurement, static and dynamic performance characteristics of measurement devices, calibration, concept of error, types and sources of error, statistical analysis of data, regression analysis, correlation, estimation of uncertainty and presentation of data, design of experiments. Sensors and Transducers: Types of sensors and transducers, Open loop and Closed loop measurement system, static characteristics i.e. accuracy, precision, sensitivity, resolution, linearity, range, etc.		
UNIT-II		11 Hours
Measurement of field quantities: Temperature, pressure, velocity by intrusive and non-intrusive methods under steady and transient conditions. Measurement of derived quantities: Flow rate, temperature in flowing fluids. Measurement of force, torque and strain: Load cells, strain gauges, pneumatic and hydraulic systems.		
UNIT-III		10 Hours
Metrology and Inspection: Types and sources of error, Controllable and Random Errors, Limit fits and tolerances, Interchangeability and standardization. Linear, angular measurements devices and Comparators: Line and End standards, slip gauges, sine bar, different types of gauges, Taylor's Principle of Gauge Design. Mechanical, electrical and pneumatic comparators.		
UNIT-IV		11 Hours
Measurement of geometric forms: Straightness, flatness, roundness, mechanical and optical methods. Interferometry: Principle and use of interferometer, optical flat, testing of parallelism of a surface with the help of optical flat. Surface texture: Different types of irregularities, Quantitative evaluation of surface roughness and texture, CMM, working and features.		
Text Books		
1	A.K. Bewoor, V.A. Kulkarni, "Metrology and Measurement", McGraw Hill Education India, First Edition, 2009	
2	I.C. Gupta, "Engineering Metrology", Dhanpat Rai Publications, 2019	
3	T.G. Beckwith, R.D. Maragoni and J.H Lienhard, "Mechanical Measurements", Pearson, Sixth Edition, 2013.	
Reference Books		
1	F.W. Galyer, C.R. Shotbolt, "Metrology for Engineers", ELBS edition, 2009	
2	R.K. Jain, "Engineering Metrology", Khanna Publishers, 22 nd Edition, 2022	
3	Czichos, "The Springer handbook of metrology and Testing", Third Edition, 2011	

SOLAR ENERGY TECHNOLOGY	
Course Code: BMA 218 Contact Hours: L-3 T-1 P-0 Course Category: Interdisciplinary	Credits: 4 Semesters: 3

Introduction: This course imparts profound knowledge in a special field such as measurement of solar radiation, importance of solar energy. The student gains specialized knowledge in the conversion of solar energy into electricity.

Prerequisite: NIL

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Understand concept of solar and different types of Solar Cells, Modules and Solar Radiation, Explain the Need, importance and scope of Solar Energy resources.

CO2: Understand and analyze the working of Solar Thermal Power Plants using single line diagram and state the functions of the major equipment and auxiliaries of each plant.

CO3: Learn the solar photovoltaic technology and storage of solar energy.

CO4: Estimate the solar power and distribution and apply the concept of solar energy in several solar appliances.

Pedagogy: Classroom teaching is supported by White board, black board, chalks, markers, projector and screen. The hand-written notes, Power Point slides and assignments will be provided to the students and also mailed to them. The students can also raise their issues related to the course in the class and mail.

Contents

UNIT-I	10 Hours
Solar Radiation: Sun as a source of energy, Solar radiation, Solar radiation at the Earth's surface, Measurement of Solar radiation - Pyroheliometer, Pyranometer, Sunshine recorder, Prediction of available solar radiation, Solar energy-Importance, Storage of solar energy, Solar pond	
UNIT-II	10Hours
Solar Thermal Systems: Principle of conversion of solar radiation into heat, Collectors used for solar thermal conversion: Flat plate collectors and Concentrating collectors, Solar Thermal Power Plant, Solar cookers, Solar hot water systems, Solar dryers, Solar Distillation, Solar greenhouses.	
UNIT-III	10 Hours
Solar Photovoltaic Systems: Conversion of Solar energy into Electricity - Photovoltaic Effect, Solar photovoltaic cell and its working principle, Different types of Solar cells, Series and parallel connections, Photovoltaic applications: Battery chargers, domestic lighting, street lighting and water pumping.	
UNIT-IV	10 Hours
Solar PV System Design and Integration: Solar Radiation Energy Measurements, Estimating Energy requirement, Types of Solar PV System, Design methodology for SPV system, Design of Off Grid Solar Power Plant, Case studies of 3KW Off Grid Solar PV Power Plant, Design and Development of Solar Street Light and Solar Lantern, Off Grid Solar power Plant.	
Text Books	
1.	G. D. Rai, Solar Energy Utilization, Khanna Publishers
2.	G.N. Tiwari, Solar Energy- Fundamentals, Design, Modeling & Applications, Narosa Pub., 2005.
3.	S.P. Sukhatme, Solar Energy-Principles of thermal energy collection & storage, Tata Mc- Graw Hill Publishers, 1999
Reference Books	
1.	Chetan Singh Solanki, Solar Photovoltaics- Fundamentals, Technologies and Applications, PHI Learning Pvt. Ltd.,
2.	P. Jayarama Reddy, Science and Technology of Photovoltaics, BS Publications, 2004
3.	D Y Goswami, F Kreith and J F Kreider, Principles of Solar Engineering, Taylor and Francis, 1999.

INDUSTRIAL WASTE WATER TREATMENT	
Course Code: BMA 219 Contact Hours: L-3 T-1 P-0 Course Category: Interdisciplinary	Credits: 4 Semesters: 3

Introduction: This course provide an understanding of the mechanisms and processes used to treat waters that have been contaminated in some way by anthropogenic industrial or commercial activities prior to its release into the environment or its re-use. To understand various terms used in industrial wastewater treatment and to acquaint with different steps involved in treatment of industrial wastewater.

Course Objective:

- The objective of the teacher is to impart knowledge and abilities to the students to:
- Distinguish between the quality of domestic and industrial water requirements and Wastewater quantity generation.
- Understand the industrial process, water utilization and waste water generation.
- Impart knowledge on selection of treatment methods for industrial waste water.
- Acquire the knowledge on operational problems of common effluent treatment plants.
- Gain knowledge on different techniques and approaches for minimizing the generation and application of Physio chemical and biological treatment methods for recovery, reuse and disposal of industrial wastewater.

Prerequisite: Environmental Science

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Understand the concept of Industrial waste and sources of pollution.

CO2: Analyze the concept of Pre and Primary waste treatment.

CO3: Evaluate the compositions of waste water and manufacturing processes of industries.

CO4: Create a model to treat the wastage from industries.

Pedagogy: Classroom teaching is supported by White board, black board, chalks, markers, projector and screen. The hand-written notes, Power Point slides and assignments will be provided to the students and also mailed to them. The students can also raise their issues related to the course in the class and mail.

Contents

UNIT-I	10 Hours
Definition: Water and Waste Water- Importance of water, water requirements (Domestic, Institutional, Industrial, Public and Agriculture) Water and Waste Water Characteristics Sources of Pollution - Physical, Chemical, Organic & Biological properties of Industrial Wastes - Difference between industrial & municipal waste waters - Effects of industrial effluents on sewers and Natural water Bodies.	
UNIT-II	10 Hours
Pre & Primary waste Treatment - Equalization, Proportioning, Neutralization, Oil separation by Floating-Waste Reduction-Volume Reduction-Strength Reduction, Waste Treatment Methods Nitrification and De-nitrification-Phosphorous removal -Heavy metal removal - Membrane Separation Process - Air Stripping and Absorption Processes - Special Treatment Methods - Disposal of Treated Waste Water.	
UNIT-III	10 Hours
Characteristics and Compositions -Characteristics and Composition of waste water and Manufacturing Processes of Industries like Sugar, Characteristics and Composition of Industries like Food Processing Industries, Steel, and Petroleum Refineries.	
UNIT-IV	10 Hours
Characteristics and Composition of Industries -Characteristics and Composition of Industries like Textiles, Tanneries, Atomic Energy Plants and other Mineral Processing Industries – Joint Treatment of Raw Industries waste water and Domestic Sewage – Common Effluent Treatment Plants(CETP) – Location, Design, Operation and Maintenance Problems – Economical aspects.	
Text Books	
1.	Metcalf & Eddy, “Wastewater engineering Treatment disposal reuse”, Tata McGraw Hill, 2002.
2.	Eckenfelder, W.W., “Industrial Water Pollution Control”, McGraw-Hill 3 rd Edition, 1999.
3.	M.N. Rao and Dutta, “Industrial Waste, Oxford, 3 rd Edition, 2017.
Reference Books	
1.	Mark J. Hammer, Mark J. Hammer, Jr., “Water & Wastewater Technology”, Prentice Hall of India, 6 th Edition, 2008.
2.	N.L. Nemerrow, “Theories and practices of Industrial Waste Engineering. Addison-Wesley Publishing Company, 1963.
3.	C.G. Gurnham, “Principles of Industrial Waste Engineering”, Wiley, 1955.

B. Tech Fourth Semester Syllabus

Computer Organization & Architecture	
Course Code: BCS- 202 Contact Hours: L-3 T-0 P-2 Course Category: DCC	Credits: 4 Semester: 4

Introduction:

In order to achieve complete understandings of computer systems, it is always important to consider both hardware and software design of various computer components. In other words, every functionality of the computer has to be studied to increase the performance of the computer. Computer organization and architecture mainly focuses on various parts of the computer in order to reduce the execution time of the program, improve the performance of each part.

Course Objective:

- Understand the basics of computer organization: structure and operation of computers and their peripherals.
- Understand basic processing unit and organization of simple processor.
- Expose different ways of communicating with I/O devices and standard I/O interfaces.
- Understand concept of pipelining and other large computing system.

Pre-requisite: Fundamentals of computers and digital logic.

Course Outcome:

- CO1: Ability to Demonstrate an understanding of the design of the functional units of a digital computer system.
- CO2: Explain the instruction set, instruction formats and Addressing modes of CPU
- CO3: Ability to Recognize and manipulate representations of numbers stored in digital computers and perform Basic arithmetic Operations.
- CO4: Ability to analyze memory hierarchy and its impact on computer Cost/performance.

Pedagogy:

Classroom teaching which focuses on developing understanding of students to digest the concepts of subject with large number of examples.



		12 Hours
UNIT-I		
Digital Logic Circuit: Basic Logic functions, Synthesis of logic functions using basic and universal gates, Boolean Algebra Properties, Flip-Flops, Registers, Shift- Registers, Counters, Decoders, Multiplexers, Functional Unit of computer system. Data Representation: Data types, R & (R-1)'s Complements, Fixed-Point representation, Floating point representation. Register Transfer and Micro operations: Register transfer language, register transfer, Bus and Memory transfer, Arithmetic Micro operations, Logic Micro operations, Shift Micro operations		
		10 Hours
UNIT-II		
Basic Computer Organisation and Design: Instruction Codes, Computer Instructions, Timing and Control, Instruction Cycle, Memory Reference Instructions, Input-Output and Interrupt. Micro programmed Control: Control Memory. Central Processing Unit: Stack Organization, Instruction Formats, Addressing Modes, Program Control, Reduced Instruction Set Computer: RISC characteristics, CISC characteristics. Performance and Metrics.		
		10 Hours
UNIT-III		
Pipelining and Vector Processing: Parallel Processing, Pipelining, Arithmetic Pipelining, Instruction Pipelining, RISC Pipelining, Vector Processing, Array Processors. Computer Arithmetic: Addition and Subtraction, Multiplication Algorithms, Division Algorithms, Floating- Point Arithmetic Operations.		
		10 Hours
UNIT-IV		
Input-Output Organization: Peripheral Devices, Input-Output interface, Asynchronous data transfer, Modes of transfer, Priority Interrupt, Direct Memory Access. Memory organization: Memory Hierarchy, Main Memory, Auxiliary Memory, Associative Memory, Cache Memory, Virtual Memory, Memory Management Hardware.		
Text Books		
1	M. Morris Mano, "Computer System Architecture", PHI, 3 rd Edition, 2016/Latest Edition.	
2	Carl Hamacher, Zvonko Vranesic, Safwat Zaky, "Computer Organization", McGraw Hill, 5 th Edition, 2012/Latest Edition.	
3	William Stallings, "Computer Organization and Architecture", PHI, 11 th edition, 2021/Latest Edition.	

Reference Books	
1.	John L. Hennessy and David A. Patterson, "Computer Architecture a quantitative approach", Elsevier, 6th Edition, 2019/Latest Edition.
2.	A. Anandkumar, "Fundamentals of digital circuits", PHI, 4th edition, 2016/Latest Edition.

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Data Communication and Computer Networks	
Course Code: BCS 204 Contact Hours: L - 3 T - 0 P - 2 Course Category: DEC	Credits: 4 Semester: 4

Introduction: This course introduces about communication protocols, layered network architectures, convention computer system interfacing standards, peer to peer data link communication protocols, Design basis network systems, and Analyze data communication technology.

Course Objectives:

- To introduce the fundamental various types of computer networks.
- To demonstrate the TCP/IP and OSI models with merits and demerits
- To explore the various layers of OSI Model.
- To introduce UDP and TCP Models.

Pre-requisite: The student should have basic knowledge of logic circuits and their applications in digital system.

Course Outcomes: After completing this course, the students will be able to:

- CO1: Remembers and understand the basics of Computer Networks and Various Protocols.
- CO2: Understand the concepts of World Wide Web Concepts.
- CO3: Administrate a network and flow of information further.
- CO4: Analyze the concepts of network security, Mobile and ad hoc networks.

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations, and quizzes. Students would be encouraged to develop an understanding of the subject. The use of ICT and web-based sources will be adopted.

CONTENTS

UNIT-I		10 Hours
Introduction: Data Communications: Components – Direction of Data flow – Networks – Components and Categories – Types of Connections – Topologies – Protocols and Standards – ISO / OSI model, Example Networks such as ATM, Frame Relay, ISDN Physical layer: Transmission modes, Multiplexing, Transmission Media, Switching, Circuit Switched Networks, Datagram Networks, Virtual Circuit Networks.		
UNIT-II		10 Hours
Stacks and Queues: Introduction, Framing, and Error – Detection and Correction, Parity – LRC – CRC Hamming code, Flow and Error Control, Noiseless Channels, Noisy Channels, HDLC, Point to Point Protocols. 111 Medium Access sub layer: ALOHA, CSMA/CD, LAN – Ethernet IEEE 802.3, IEEE 802.5 – IEEE 802.11, Random access, Controlled access, Channelization.		
UNIT-III		8 Hours
Network layer: Logical Addressing, Internetworking, Tunneling, Address mapping, ICMP, IGMP, Forwarding, Uni-Cast Routing Protocols, Multicast Routing Protocols.		
UNIT-IV		12 Hours
Transport Layer: Process to Process Delivery, UDP and TCP protocols, Data Traffic, Congestion, Congestion Control, QoS, Integrated Services, Differentiated Services, QoS in Switched Networks. Application Layer: Domain name space, DNS in internet, electronic mail, SMTP, FTP, WWW, HTTP, SNMP.		
Text Books		
1	Data Communications and Networking, Behrouz A. Forouzan , Fourth Edition TMH, 2006.	
2	Computer Networks, Andrew S Tanenbaum, 4th Edition. Pearson Education, PHI.	
3	Data communications and Computer Networks, P.C .Gupta, Second Edition Prakash C. Gupta ©. 2014 by PHI Learning Private Limited, Delhi.	
Reference Books		
1	An Engineering Approach to Computer Networks, S. Keshav, 2nd Edition, Pearson Education.	
2	Understanding communications and Networks, 3rd Edition, W.A. Shay, Cengage Learning.	
3	Computer Networking: A Top-Down Approach-Featuring the Internet- James F. Kurose & Keith W. Ross, 3 rd Edition, Pearson Education.	
4	Data and Computer Communication, William Stallings, Sixth Edition, Pearson Education, 2000.	

LIST OF PRACTICALS

S. No.	Name of the Experiment
1.	Generate a Time Division Multiplexing (TDM) signal using relevant simulation software.
2.	Create a Hybrid Network Using Bluetooth.
3.	Locate the error bit in the given data string by applying checksum error detection method.
4	Implement Wireless network.
5	Write a 'C' program for parity check error detection.
6	Write a 'C' program for Cyclic Redundancy Check (CRC) error detection.
7	Write a 'C' program for error correction using Hamming code.
8	Configure static IP address in operating system along with appropriate subnet mask for given problem.
9	Execute TCP/IP network commands: ipconfig, ping, tracert.
10	Execute TCP/IP network commands: netstat, pathping, route.
11	Install Wireshark and configure as packet sniffer- i) Capture IP, TELNET, FTP packets using Wireshark.
12	Capture TCP and UDP packet using Wireshark.
13	Capture ARP and ICMP packet Traffic using Wireshark.
14	Install Operating System Linux/Windows/Any other Server.
15	Use FTP protocol to transfer file from one system to another system.
16	Create IPv6 environment in a small network using simulator.
17	Configure TELNET for remote login.

DESIGN AND ANALYSIS OF ALGORITHMS	
Course Code: BCS 201	Credits: 4
Contact Hours: L-3 T-0 P-2	Semester: 3
Course Category: DCC	

Introduction:

This course deals with teaching different methodologies of designing algorithms. There are certain standard approaches of analyzing the algorithms. This course deals with all aspects of these analysis. It teaches the concepts of Dynamic programming, different approaches of algorithm design like Greedy approach etc.

Course Objectives:

- Introduction, learning and analysis of performances of algorithmic efficiency of approaches such as searching, sorting etc.
- Introduction, learning and analysis of greedy paradigms.
- Introduction, learning and analysis of dynamic programming and back tracking.
- Introduction, learning and analysis of computational complexity and branch & bound.

Pre-requisites: Data structures.

Course Outcomes: After completion of the course, the students will be able to:

CO1: Understand asymptotic complexities of the algorithms and design algorithms using Divide and Conquer approach.

CO2: Understand and apply greedy and dynamic programming approaches for designing algorithms.

CO3: Understand, analyse and implement various graph algorithms and the backtracking approach of algorithm design.

CO4: Understand and implement different string-matching algorithms and NP-Complete problems.

Pedagogy:

Classroom teaching which focuses on developing understanding of students to digest the concepts of subject with large number of examples.

Contents

UNIT-I	10 Hours
Introduction: Algorithm definition and specification, analysis of algorithmic efficiency of algorithms Review of growth of function, space complexity, time complexity, Recurrences: Substitution method, Iteration method, Master method Divide and Conquer Approach: binary search, merge Sort, quick sort, shell sort, heap-sort, Simultaneous Max and Min Problem, Strassen's algorithm for matrix multiplications.	

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UNIT-II		10 Hours
Greedy Algorithms: Elements of Greedy strategy, knapsack problem, job sequencing with deadlines, Activity selection problem, Huffman Codes. Dynamic Programming: Elements of Dynamic Programming, Matrix Chain Multiplication, Longest common subsequence and optimal binary search trees problems.		
UNIT-III		12 Hours
Disjoint set operations, union and find algorithms, Graph Algorithms: DFS, BFS, Topological Sort, Strongly Connected Components, Kruskal's and Prim's algorithm for MST, Dijkstra's and Bellman Ford Algorithm, All pair shortest paths Algorithm. Back Tracking: General method, n-queen's problem, Branch and Bound: General Method, 0/1 knapsack.		
UNIT-IV		10 Hours
String matching: Naïve String Matching algorithm, Rabin-Karp Algorithm, String Matching with finite automata, The Knuth-Morris Pratt algorithm. NP-Complete Problem: Polynomial time verification, NP-Completeness and Reducibility, NP-Completeness Proof, NP-Complete problems.		
Text Books		
1	.H .Cormen, C .E .Leiserson, R .L .Rivest, "Introduction to Algorithms", 3rd Ed., PHI/Latest Edition.	
2	E. Hörowitz, S. Sahni, and S. Rajsekaran, "Fundamentals of Computer Algorithms," 2nd Ed., Universities Press/Latest Edition.	
3	P. H. Dave, H. B. Dave, "Design and Analysis of Algorithms", 2nd Ed./Latest Edition, Pearson Education.	
Reference Books		
1	Design and Analysis of Algorithms, S. Sridhar, Oxford Univ. Press. Latest Edition.	
2	Design and Analysis of algorithms, Aho, Ullman and Hopcroft, Pearson Education, 2008./Latest Edition.	
3	Foundations of Algorithms, R. Neapolitan and K. Naimipour, 4th edition, Jones and Bartlett Student edition/Latest Edition.	

Suggested List of Experiments:

1. Merge Sort
2. Quick Sort
3. Simultaneous Min_Max
4. Strassen's Matrix Multiplication
5. Implement Disjoint Set Data Structure.
6. Job Sequencing with Deadline
7. Greedy algorithm for fractional Knapsack Problem
8. Huffman Coding Problem
9. Kruskal Algorithm
10. Prim's Algorithm
11. Dijkstra's Algorithm
12. Bellman- Ford Algorithm
13. Matrix Chain Multiplication Problem
14. All Pair Shortest Path Problem

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15. Longest Common Subsequence Problem
16. Back tracking algorithm for N – Queen Problem
17. BRANCH & BOUND algorithm for 0/1 Knapsack problem
18. Naïve String Matching Algorithm
19. Rabin Karp Algorithm
20. KMP String Matching Algorithm

FUNDAMENTALS OF ANALOG & DIGITAL ELECTRONICS

Course Code: BEC 207

Contact Hours: L-3 T-0 P-2

Course Category: IDC

Credits: 4

Semester: 3

Introduction: The course will introduce fundamental principles of analog and digital electronics. The course provides sufficient basic knowledge for the undergraduate to understand the design of diodes and transistor-based circuits, op-amps and their applications as well as the design of digital circuits.

Course Objective:

- Understand the design and analysis of various analog electronic circuits
- Understand the fundamental concepts and techniques used in digital electronics

Pre-requisite: BEC-101 Basics of Electrical and Electronics Engineering.

Course Outcome: After completion of the course, student will be able to:

CO1: Understand basic electronic devices such as diodes, BJT & FET transistors

CO2: Understand various applications of Op-Amp

CO3: Analyze logic processes and implement logical operations using combinational logic circuits

CO4: Design sequential circuits

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped classroom teaching will be adopted.

Contents

UNIT-I	10 Hours
Overview of BJT: operating Point, AC/DC Load line, leakage currents, saturation and cut-off mode of operations.	
Bias stabilization: Need for stabilization, various biasing schemes, bias stability with respect to variations in I_{co} , V_{BE} & β , Stabilization factors, thermal stability.	
BJT Amplifier Models: Low-frequency models for the transistor (h-parameter, Hybrid – Π , r_{π})	
BJT amplifiers: Cascaded Amplifiers, Darlington pair, Cascode amplifiers.	
UNIT-II	10 Hours
Multistage Amplifiers: Methods of coupling, RC coupled amplifier,	
Feedback Amplifiers: Feedback concept, Classification of Feedback amplifiers, Properties of negative feedback amplifiers.	
Overview of Field Effect Transistor: Classification, JFET characteristics, Operating point, Enhancement & depletion type MOSFETs, JFET Model, JFET amplifier analysis (CD, CS & CG)	
Introduction to Op-Amp: Op-Amp models (Ideal & Practical), Inverting & non-inverting amplifier, Transfer characteristics, Current mirror, Level Shifter, Compensating techniques.	
UNIT-III	10 Hours
Analog & Digital signals, AND, OR, NOT, NAND, NOR & XOR gates, Boolean algebra.	
Standard representation of Logical functions, K-map representation and simplification of logical functions, Don't care conditions, X-OR & X-NOR simplification of K-maps. Combinational circuits: Multiplexers, demultiplexers, Decoders & Encoders, Adders & Subtractor, Code Converters, comparators, decoder/ drivers for display devices	
Logic Implementations using combinational circuits: ROM, PAL & PLA	

UNIT-IV		10 Hours
Synchronous Sequential Circuits: Finite-state machines, latches and flip-flops (SR, D, JK, T), synthesis of clocked sequential circuits, Mealy and Moore machines.		
Registers and Counters: Registers and shift registers, Ripple counters, Synchronous counters, Counters with unused states, Ring counter, Johnson counter, sequence generators and sequence detectors.		
Text Books		
1	Morris Mano, "Digital Design", PHI, 5 th edition, 2013 (latest edition).	
2	Millman and Halkias, "Electronic Devices and Circuits" TMH, 4 th Edition, 2015 (latest edition).	
3	Salivahanan, Suresh Kumar, Vallavaraj, "Electronic Devices and Circuits" TMH, 4 th Edition, 2016 (latest edition).	
Reference Books		
1	Balbir Kumar and S. B. Jain, "Electronic Devices and Circuits" PHI, 2 nd Edition 2014 (latest edition).	
2	R.P. Jain, "Modern Digital Electronics", TMH, 4 th Edition, 2010 (latest edition).	

List of Experiments

1. To design a Common Base Configuration Circuit using NPN/PNP Transistor and plot the input and output configuration curve.
2. To design a Common Emitter Configuration Circuit using NPN/PNP Transistor and plot the input and output configuration curve.
3. To design a Half-Wave and Full-Wave Rectifier Circuit and plot the input and output characteristics response.
4. To design a Bridge Rectifier Circuit and plot the input and output characteristics response.
5. To plot frequency response of common source FET amplifier.
6. To study the application of 741 Op-Amp IC as an Inverting and Non-Inverting Amplifier.
7. To analyze the functions of a BCD-to-7-segment decoder (IC 7447) and use it to control the operations of a common-anode 7-segment LED display.
8. Analysis and Synthesis of Boolean Expressions using Basic Logic Gates.
9. Analysis and Synthesis of Logic Functions using Multiplexers.
10. Analysis and Synthesis of Logic Functions using Decoders.
11. Analysis and Synthesis of Boolean Relations using Digital Comparators.
12. Analysis and Synthesis of Arithmetic Expressions using Adders/Subtractors.
13. Analysis and Synthesis of Multi-bit Sequential Circuits using Shift Registers.
14. Analysis and Synthesis of Sequential Circuits using Basic Flip Flops.

INTRODUCTION TO IOT AND APPLICATION DESIGN CONCEPTS	
Course Code: BCS 202 Contact Hours: L: 2 T: 0 P:4 Course Category: SEC	Credits: 4 Semesters: 3

Introduction:

The Internet of Things (IoT) is a groundbreaking technological advancement that has gained significant traction with the proliferation of connected devices, sensor networks, and enhanced computing capabilities. This course is designed to introduce a broad spectrum of students to the field of IoT, offering a comprehensive and practical learning experience. It provides in-depth content that bridges theoretical concepts with real-world applications. Beginning with a fundamental introduction to IoT, the course will guide students through a detailed case study of an IoT solution, ensuring a hands-on understanding of the subject.

Course Objective:

This course focuses on developing a thorough understanding of the Internet of Things (IoT), including its architecture, application development for real-world problem-solving, network and IoT protocols, sensor and I/O device interfacing, data processing, and AI-driven IoT applications.

Prerequisite:

Students should have prior knowledge of computer fundamentals, Knowledge of at least one programming language.

Course Outcomes:

After successful completion of the course, the student will be able to:

- CO1: Understand the architecture and applications development of IoT in real-world scenario.
- CO2: Design of IoT applications by interfacing of various sensors, I/O devices and I/O peripherals
- CO3: Develop of IoT applications on different platforms.
- CO4: Deployment of IoT applications on different platforms in in real-world scenario.

Pedagogy:

The course will be delivered through a combination of lectures, tutorials, assignments, projects, presentations, and quizzes. Faculty members aim to create an interactive learning environment, enabling students to connect theoretical concepts with real-world applications for deeper understanding. Additionally, Information and Communication Technology (ICT), web-based resources, and flipped classroom methodologies will be integrated to enhance the teaching-learning experience.

Contents

UNIT-I		7 Hours
IoT Introduction: Introduction and Definitions of Internet of Things (IoT), IoT growth-a statistical view, functional requirements, characteristics, foundations, architectures, challenges and issues. Physical design of IoT, Logical design of IoT, Application areas of IoT in Real world scenarios. Enabling technologies: IoT Building Blocks, Sensor, Actuators, Computing Platforms, Communication protocols and Programming Environments.		
UNIT-II		7 Hours
IoT enabling platforms and programming language: Detail hardware architecture of Arduino, Esp32, 8051 target IoT platforms. Programming Languages for IoT: Introduction to Embedded C and Arduino Programming, Its IDEs, Tools Chain, Project Building, Debugging and testing. Basic Programs with the selected hardware platforms.		
UNIT-III		7 Hours
Interfacing: Interfacing and programming of various sensors, Input and output peripherals for IoT Application Design and data generation. Interfacing and programming of various IOT communication Technologies: RFID, Bluetooth, Zigbee, Wi-fi Connectivity. Design of IOT Application integration of Technologies: Databases Connectivity over data bases, development of Web Applications/ Mobile Applications, Cloud, Storage in Cloud and Data visualisation.		
UNIT-IV		7 Hours
Domain specific applications IoT application design and development with case studies: Home automation, Industry applications, Surveillance applications, Smart Homes, Ambient Assisted Living, Intelligent Transport. Visualization and interpretation of Data, Data Cleaning, Testing and deployment of IoT and Project (Mini Project).		
Text Books		
1.	R. Kamal, "Internet of Things: Architecture and Design Principles", 1 st Edition, McGraw Hill Education private limited, 2017/ Latest Edition.	
2.	S.R.N. Reddy, "Embedded Systems & IoT design and development(Using 8051 Microcontroller)", 1 st Edition, EduTech Labs	
3.	A. Bahga, V. Madiseti, "Internet of Things: A Hands-on Approach", 1 st Edition, Universities Press, 2015/ Latest Edition.	
Reference Books		
1.	Surya Durbha, Jyoti Joglekar, Internet of Things, Oxford University Press 2019	
2.	O. Hersent, D. Boswarthick, O. Elloumi, "The Internet of Things – Key applications and Protocols", 2nd Edition, Wiley, 2012/ Latest Edition.	
3.	Andrew Minter, Analytics for the Internet of Things (IoT), Packt Publications, Jul 2017	

List of Practical's:

1. Introduction to programming language (Embedded C) and programming development environments
2. Introduction to IoT application development with selected hardware platforms (eg Arduino)
3. Interfacing LED with hardware platform and blinking it with 2 sec intervals.
4. Interfacing the digital sensors (IR, PIR and Touch sensor, Light sensor) and reading sensor data.
5. Interfacing temperature and humidity sensor with target platform and display the sensor data.
6. Using the same board as in (1), read data from a sensor. Experiment with both analog and digital
7. Interfacing soil moisture sensor with target hardware platform and display the alarming notification to user.
8. Design and development of traffic light control system on the target platform.
9. Design and development of temperature control system using target platform
10. Interfacing actuators and activation of actuators as sensor's data cross the threshold.
11. Interfacing Bluetooth with target platform and sending data to mobile phone.
12. Mini Project

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SEMICONDUCTOR TECHNOLOGY	
Course Code: B EC 208	Credits: 4
Contact Hours: L-3 T-0 P-2	Semester: 3
Course Category: IDC	

Introduction: The course will introduce concepts of semiconductor devices which form the basic building block of Chip design and VLSI circuits. It will also give insight on IC fabrication process and complete flow of IC design using CAD tools. Industry standard tools for both front end and backend design will be introduced at introductory level only. The course also deals in latest trends in semiconductor technology.

Course Objective:

- To give an insight into fundamental concepts of semiconductor devices used in VLSI circuits
- To give the broad spectrum of fabrication techniques for integrated circuits and Visualize the basic semiconductor device fabrication process steps
- To gain insight into complete chip design flow and tools used.
- To study recent and emerging trends in semiconductor devices and VLSI techniques.
- Qualitatively explain the possible reliability challenges of the emerging devices and their possible solutions

Pre-requisite:

- Basic knowledge of Theory of semiconductor physics, electronic devices, digital circuits and programming concept.

Course Outcome: After completion of the course, student will be able to:

CO1: Understand MOS transistor-based circuits and implementation of circuits using MOS Transistor

CO2: Understand the fundamental principles of semiconductor fabrication and packaging and its significance in the VLSI industry.

CO3: Gain proficiency in using industry-standard software tools for chip design simulation and FPGA boards

CO4: Apply knowledge of semiconductor technology to real-world applications and industry challenges

Pedagogy: Class room teaching, tutorials, Practical based learning

Contents

UNIT-I	10 Hours
Semiconductor Devices for VLSI- MOS Transistor and its operating regions, CMOS Inverter, VI Characteristics, Noise margins and power dissipation, Combinational circuit implementation using MOS transistor, NAND, NOR design etc, Sequential circuit implementation using MOS transistor, Flip Flop, Register Design etc	
UNIT-II	12 Hours
IC Fabrication: Crystal growth, clean rooms, Wafer cleaning, Device isolation, Well formation, Gate and source/drain formation, Contact and local interconnects, Multilevel	

metal formation NMOS, PMOS and CMOS Fabrication, Packaging techniques, Through hole, Surface mount, Bond wire, Flip chip technology
Introduction to Layout and Foundary, TCAD; Role of TCAD in semiconductor device development.

UNIT-III

10 Hours

Chip Design and Simulation: SPICE programming, Introduction to IC design tools (Cadence/Mentor Graphics tool etc) Introduction to physical design automation, Partitioning, Floor planning, placement, routing, clock and timing.
Introduction to HDL programming, testbench and simulation, Xilinx Tool and Industry standards, FPGA

UNIT-IV

10 Hours

Emerging trends in semiconductor technology: Requirements of high-performance nanoscale devices, Silicon-on insulator, Multi-gate Transistors, FinFET, Gate-all-around, Ferroelectric Transistors and Stacked Transistors, Carbon nano tubes, Case study of new devices (Complementary FET, Nanowire) and industry standards for Chip Design

Text Books

- 1 James D. Plummer, M.D. Deal and P.B.Griffin, "Silicon VLSI Technology, Fundamentals, Practice and Modeling", 1st Edition, Pearson Education, 2009.
- 2 C. Unsala and B. Tar, Digital System Design with FPGA: Implementation Using Verilog and VHDL, McGraw Hill, 2017
- 3 Colinge J. P., "FinFET and Other multi-gate-transistor, 2008, Springer, ISBN 978-0-387-71751-7
- 4 Yoon S. M, "Ferroelectric gate Field effect transistor, device physics and application, 2020, Springer
- 5 M. S. Tyagi, Introduction to Semiconductor Materials and Devices, 1st ed., 1991, John Wiley Sons, New York, USA, ISBN 978-8126518678

Reference Books

- 1 S.M.Sze, "VLSI Technology", 2nd Edition, McGraw-Hill, 2017.
- 2 Sedra & Smith, "Micro Electronic Circuits" Oxford University Press, 6th Edition, 2012 (latest edition).
- 3 Richard C. Dorf, John V. Oldfield, Field-Programmable Gate Arrays: Reconfigurable Logic for Rapid Prototyping and Implementation of Digital Systems, Wiley, 2008
- 4 C. Unsalan, B. Tar, Digital system design with FPGA: Implementation using Verilog and VHDL, TMH, 2017

Topic 1. To identify various components being used in any electronic circuit such as resistor, capacitor, various diodes (p-n junction, Zener, LED), transistors (BJT, MOSFET, FET), breadboard, potentiometer.

- Learn graphical symbols used to represent the various components.
- Find the value of resistance, capacitance by its color code and value mentioned on the component.

Topic 2. To study the various control on the panel of a typical DSO, Multimeter.

Topic 3. To study the various control on the panel of a function generator and DC power supply.

Topic 4. To Study the pin configuration of a given IC number.

Topic 5. To Measure the current, voltage & Power using electrical/electronic instruments

Topic 6. To study and verify the truth table of various logic gates on the breadboard.

Topic 8. To design an electronic circuit with discrete components on Breadboard/general purpose PCB.

Topic 9. To realize schematic of an electronic circuit using any design software (OrCAD/TINA/ KiCAD/ DesignSpark PCB/ any other available software).

Topic 10. To identify various peripheral devices including a keyboard, mouse, printer, and flash drive of a computer

Topic 11. To study motherboard and various ports in a computer for interaction with computer and identify various hardware peripherals like RAM, ROM and Processor

NUMERICAL METHODS

Course Code: BAS- 205

Contact Hours: L-3 T-1 P-0

Course Category: IDC

Credits: 4

Semester: 3/4

Introduction: Numerical Methods give insight into problems we cannot otherwise solve. These methods provide us the way to solve problem when exact methods fails or unable to produce the desirable results.

Course Objectives:

- To motivate the students to understand and learn various numerical techniques to solve mathematical problems representing various engineering, physical and real life problems.
- To provide constructive methods for obtaining answers to such problem for which analytical methods fails to find solutions.

Pre-requisites: Calculus, Differential equations, some exposure to linear algebra (matrices).

Course Outcomes: Upon completion of this course, the students will be able to:

CO1: Understand the errors, sources of error and its effect on any numerical computations and also analyze the efficiency of any numerical algorithms.

CO2: Learn how to obtain numerical solution of nonlinear equations using bisection, secant, Newton, and fixed-point iteration methods.

CO3: Solve system of linear equations numerically using direct and iterative methods.

CO4: Understand how to approximate the functions using interpolating polynomials.

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects, presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped classroom teaching will be adopted.



Content

UNIT-I	10 Hours
Floating-Point Numbers: Floating-point representation, rounding, chopping, error analysis, conditioning and stability. Non-Linear Equations: Bisection, secant, fixed-point iteration, Newton method for simple and multiple roots, their convergence analysis and order of convergence.	
UNIT-II	11 Hours
Linear Systems and Eigen-Values: Gauss elimination method using pivoting strategies, LU decomposition, Gauss-Seidel and successive-over-relaxation (SOR) iteration methods and their convergence, ill and well- conditioned systems, Rayleigh's power method for eigen-values and eigen-vectors.	
UNIT-III	11 Hours
Interpolation and Approximations: Finite differences, Newton's forward and backward interpolation, Lagrange and Newton's divided difference interpolation formulas with error analysis, least square approximations. Numerical Integration: Newton-Cotes quadrature formulae (Trapezoidal and Simpson's rules) and their error analysis, Gauss--Legendre quadrature formulae.	
UNIT-IV	10 Hours
Differential Equations: Solution of initial value problems using Picard, Taylor series, Euler's and Runge- Kutta methods (up to fourth-order), system of first-order differential equations.	
Text Books	
1	Jain M.K., Iyengar, S.R.K., and Jain, R.K., Numerical Methods for Scientific and Engineering Computation, 6 th Edition, New Age International Publication, 2012 (Latest edition).
2	Sastry S., Introductory Methods of Numerical Analysis, 5 th Edition, Prentice Hall India Learning Private Limited; 2012.
3	Conte, S.D and Carl D. Boor, Elementary Numerical Analysis: An Algorithmic Approach, SIAM-Society for Industrial and Applied Mathematics, 2017.
4	Grewal, B. S., Higher Engineering Mathematics, 44 th Edition, Khanna Publishers, 2012.
Reference Books	
1	Gerald C.F and Wheatley P.O., Applied Numerical Analysis, 8 th Edition, Pearson Education, 2011.
2	Chappra S.C., Numerical Methods for Engineers, 7 th Edition, McGraw-Hill Higher Education, 2014.





OPTIMIZATION TECHNIQUES	
Course Code: BAS-206	Credits: 4
Contact Hours: L-3 T-1 P-0	Semester: 3/4
Course Category: IDC	

Introduction: Having a sound foundation of applied Mathematics; students are well equipped to apply them in various fields including Optimization Techniques which provides a logical and systematic approach for decision making.

Course Objective:

- To formulate mathematical models and to understand solution methods for real life optimal decision problems.
- To emphasize the basic study of linear programming problem, Integer programming problem, Transportation problem, Two person zero sum games with economic applications and project management techniques using PERT and CPM.

Prerequisite: A basic course in calculus and matrices.

Course Outcomes (CO)

Having successfully completed this course, the student will be able to

- CO 1. Understand the concepts of Linear Space, formulation of real world problems as linear programming model and its solution using Simplex Algorithm.
 - CO 2. Demonstrate Revised Simplex Method, investigate sensitivity analysis, introduce and apply duality theory in linear programming problems.
 - CO 3. Introduce and demonstrate algorithms such as Transportation and Assignment problems and other allocation problems.
 - CO 4. Execute project management problems using CPM and to identify decision strategies of a two-person zero sum game problem arising in a competitive environment.
- **Pedagogy:** Apart from class room teaching, main focus is to enhance problem solving ability supported by weekly assignments and discussing individual's doubts.



Content

UNIT-I	12 Hours
Formulation of Linear Programming Problem (LPP), convex set, Graphical method, LPP in standard form, Solution of LPP by simplex method, Big – M Method, Two Phase Method, Exceptional cases in LPP.	
UNIT-II	10 Hours
Revised Simplex Method, Duality theory, Dual simplex method, Sensitivity analysis, Integer Programming: Branch and bound technique, Gomory's Cutting Plane Method.	
UNIT-III	10 Hours
Transportation and Assignment Problem : Initial basic feasible solutions of balanced and unbalanced transportation/assignment problems and their optimal solutions, Transshipment, Travelling Salesman Problem	
UNIT-IV	10 Hours
Game Theory: Two person zero-sum game, Game with mixed strategies, Graphical method and solution by linear programming. Project Management: Construction of networks, Network computations, Floats (free floats and total floats), Critical path method (CPM), Crashing.	
Text Books	
1	Kambo N. S., Mathematical Programming Techniques, East-West Press Pvt. Ltd., 2005
2	Chandra S., Jayadeva, Aparna Mehra, Numerical Optimization with Applications, Narosa Publishing House, 2009
Reference Books	
1	Gilbert Strang, Linear Algebra and its Applications, Cengage Learning, 4 th ed. 2010.
2	Taha H.A., Operations Research-An Introduction, PHI (2007).
3	Pant J. C., Introduction to optimization: Operations Research, Jain Brothers (2004)
4	Bazaarra Mokhtar S., Jarvis John J. and Shirali Hanif D., Linear Programming and Network flows, John Wiley and Sons (1990)
5	Ravindran, A., Phillips, D.T. and Solberg, J.J., Operations Research: Principles and Practice, John Wiley and Sons, NY, Second Edition (Reprint).
6	Ritu Arora and Kavita Gupta, Linear Programming and Game Theory, Narosa Publishing House (2022).





MATERIAL SCIENCE AND ENGINEERING	
Course Code: BAS - 207	Credits: 04
Contact Hours L: 3- T: 1 - P: 0	Semester: 3/4
Course Category:	

Introduction: Materials Science and Engineering is an interdisciplinary subject rooted in applied physics that studies the relationship between structure, properties, processing, and performance of materials. This course provides fundamental knowledge of electronic, magnetic, dielectric, optical, and modern engineering materials. It also introduces essential characterization techniques used to analyze material structure and properties. The course emphasizes the role of materials in modern electronic, photonic, and energy devices.

Course Objectives: The course aims to-

- Provide fundamental understanding of classification, structure, and properties of materials from a physics perspective.
- Explain electronic, dielectric, magnetic, and superconducting properties of materials and their applications.
- Introduce modern functional materials such as photonic materials, smart materials, nanomaterials, and biomaterials.
- Familiarize students with basic principles and applications of material characterization techniques.
- Develop the ability to correlate material properties with engineering and device applications.

Pre-requisites: Basic understanding of Applied Physics Course.

Course Outcomes: Upon completion of this course, the students will be able to:

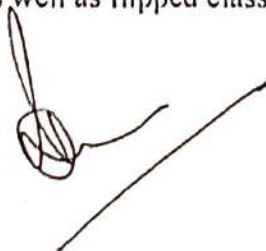
CO1: Classify materials and explain their structural and electronic properties.

CO2: Describe dielectric, magnetic, and superconducting properties and their technological applications.

CO3: Understand working principles and applications of modern materials including photonic, smart, nano, and biomaterials.

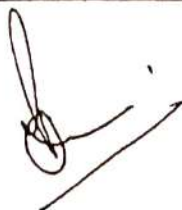
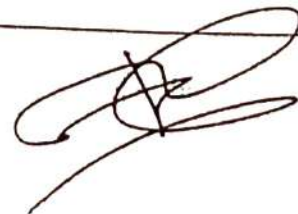
CO4: Explain fundamental principles and applications of material characterization techniques such as XRD, SEM, and spectroscopy.

Pedagogy: The teaching-learning of the course would be organized through lectures, assignments, tutorial sessions (problem solving, numerical exercises and concept discussions), projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based sources as well as flipped class room teaching will be adopted.




Contents

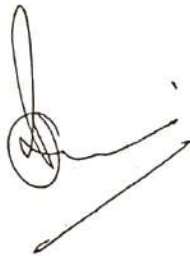
UNIT-I Electronic Materials	10 Hours
Band Theory and Materials Classification: Energy band formation in solids; classification of materials as conductors, semiconductors, and insulators based on band gap. Semiconductors and Charge Transport: Intrinsic and extrinsic semiconductors, carrier concentration, and Fermi level variation. Electrical conductivity and mobility in semiconductors. Hall Effect and Applications: Hall effect and its use in determining carrier type and concentration. Applications in Hall sensors (used in automobiles and smartphones) Applications: Role of semiconductor materials in electronic devices such as diodes, transistors and sensors.	
UNIT-II Dielectric, Magnetic, and Superconducting Materials	11 Hours
Dielectric Materials: Dielectric polarization mechanisms (electronic, ionic, orientational, space charge). Dielectric constant, dielectric loss, frequency dependence. Piezoelectric materials and applications. Magnetic Materials: Origin of magnetism, magnetic dipole moment. Types of magnetism: diamagnetism, paramagnetism, ferromagnetism, antiferromagnetism, ferrimagnetism. Magnetic hysteresis, soft and hard magnetic materials, applications. Superconducting Materials: Properties of superconductors, Meissner effect, critical temperature, Type I and Type II superconductors.	
UNIT-III Modern Engineering Materials	11 Hours
Photonic Materials: Optical properties of materials, absorption, emission, photoconductivity. LED and photodetectors: principle, working, and applications. Smart Materials: Introduction and applications of shape memory alloys, piezoelectric materials, and chromic materials. Nanomaterials: Introduction, size-dependent properties, quantum confinement, applications in electronics and energy devices. Biomaterials: Introduction, properties, and applications. Metallic biomaterials (stainless steel, titanium alloys). Polymeric biomaterials and biomedical applications.	
UNIT-IV Materials Characterization Techniques	10 Hours
Structural and Morphological Characterization: X-ray diffraction (XRD): Bragg's law, principle, and applications. Electron microscopy: SEM and TEM – principle and applications.	

Optical Characterization: UV-Visible spectroscopy: principle and applications. FTIR spectroscopy: principle and applications.

Text Books

1. William D. Callister, Materials Science and Engineering: An Introduction, 8th Edition, John Wiley & Sons, 2010.
2. Charles Kittel, Introduction to Solid State Physics, Wiley.
3. Sam Zhang, Lin Li, Ashok Kumar, "Materials Characterization Techniques", 1st Edition, CRC Press, 2008
4. T. Pradeep, "A Text Book of Nanoscience and Nanotechnology", Tata McGraw Hill, New Delhi, 2012.
5. Elements of X-ray Diffraction, B. D. Cullity, S.R. Stock, 3rd Edition, Pearson, 2001
6. R. F. Egerton, Physical Principles of Electron Microscopy: An Introduction to TEM, SEM, and AEM, 2nd Edition, Springer, 2016.



ADVANCED IOT AND REAL WORLD APPLICATIONS

Course Code: BCS 205
Contact Hours: L-2 T-0 P-4
Course Category: SEC

Credits: 4
Semesters: Fourth

Introduction:

The Internet of Things (IoT) represents a transformative paradigm in modern technology, driven by the rapid expansion of interconnected devices, intelligent sensor networks, and advanced computational frameworks. This course delves into the complexities of IoT architectures, equipping students with both theoretical insights and hands-on expertise in designing, deploying, and optimizing IoT solutions. Covering advanced IoT structures, the curriculum integrates real-world case studies to demonstrate system interoperability, edge computing, AI-driven analytics, and secure data transmission. Through a blend of conceptual learning and practical implementation, students will gain a deep understanding of IoT ecosystems and their applications in solving complex technological challenges.

Course Objective:

This course focuses on developing a advanced complex IoT structures of the Internet of Things (IoT), including its real word applications case studies, design and development complex IoT architecture, advanced application development for AIoT, IIoT, HIIoT, embedding network and IoT protocols architectures, cloud storage and IoT operating systems, data generation, storage and visualization for IoT applications.

Prerequisite:

Students should have prior knowledge of computer fundamentals, Knowledge of at least one programming language.

Course Outcomes:

After successful completion of the course, the student will be able to:

CO1: Understand the architecture and applications development of IoT for high-end real-world scenarios.

CO2: Design of high-end real-world IoT applications by interfacing of various sensors, I/O devices and I/O peripherals with high-end computing platforms

CO3: Develop Protocols and Real time operating systems for specific IoT Applications.

CO4: Design and deployment of AIoT, IIoT, HIIoT and other real world IoT based Applications

Pedagogy:

The course will be delivered through a combination of lectures, tutorials, assignments, projects, presentations, and quizzes. Faculty members aim to create an interactive learning environment, enabling students to connect theoretical concepts with real-world applications for deeper understanding. Additionally, Information and Communication Technology (ICT), web-based

S.No.	List of Experiments
1.	Verify truth tables of AND, OR, NOT, NAND, NOR, XOR and XNOR gates.
2.	Verify that NAND and NOR gates are universal gates.
3.	Design a Half adder circuit and verify it's truth table.
4.	Design a Half subtractor circuit and verify it's truth table.
5.	Design a Full adder circuit and verify it's truth table.
6.	Design a full subtractor circuit using half subtractor and verify its truth table.
7.	Design a 2-bit binary incrementor using half adder circuit.
8.	Design and implement a 3x 8 Decoder.
9.	Design and implement a 4x 1 Multiplexer.
10.	Design and implement SR flip flop
11.	Design and implement D flip flop
12.	Design and implement JK flip flop
13.	Design and implement T flip flop
14.	Design and implement Master Slave flip flop









resources, and flipped classroom methodologies will be integrated to enhance the teaching-learning experience.

Contents

UNIT-I	7 Hours
Introduction to IoT Application Design Principals: IoT Design Principles and needed Capabilities- Sensing and Actuation. Basics of Networking- Devices and gateways, Data Management, M2M and IoT. Business processes in IoT- Everything as a Service (XaaS). IoT Enabling Computing Platforms: Detail hardware architecture of RPI, JetsonNano, STM 32 Target Platforms for IoT and its salient features. Programming Languages for advanced IoT: Introduction to Python, File Handling, Python Packages for IoT, IoT Physical Servers – Communication APIs, RESTful Web API. Introduction to Embedded C and IDES[Cube IDE], Tools Chain, Project Building, Debugging and testing.	
UNIT-II	7 Hours
IoT Protocols and Cloud Storage: IoT protocols- Introduction, Message Queuing Telemetry Transport (MQTT), Constrained Application Protocol(CoAP), Extensible Messaging and Presence Protocol (XMPP), Data Distribution Service(DDS), Bluetooth Low Energy, Light Fidelity(Li – Fi), IPv4, IPv6, Internet Protocol Version 5(IPv5), Uniform Resource Identifier (URI). Introduction to IoT Cloud- and Fog Computing: Insights into ThingSpeak Cloud, Relational DBMS in Cloud, Non- Relational DBMS in Cloud. Domain Specific IoTs- Home. City, Environment, Energy, Agriculture and Industry.	
UNIT-III	7 Hours
IoT Operating System: Introduction to FreeRTOS/RTLinux, Pre-emptive and Non-preemptive scheduling, Interrupt Handling, RTLinux (Scheduling, Interrupts Handling), Task Management, Resource Management, Memory Management. IoT Physical Devices and Endpoints: Introduction to STM32, Jetson Nano and Raspberry Pi- Installation, Interfaces(Serial. SPI, I2C) Controlling Hardware- Connecting LED, Buzzer, switching High Power Devices with transistors, Controlling AC Power devices with Relays, Controlling Servo Motor, Speed Control of DC Motor, Unipolar and Bipolar Stepper Motors IoT Security: Attack, Defense, and Network Robustness of IoT, Malware Propagation and Control in Internet of Things Privacy Preservation Data Dissemination, Trust and Trust Models for the IoT, Authentication in IoT. Computational Security for the IoT Security Protocols for IoT Access Networks, Security Testing, Integration of IoT Cloud Platforms, Things Speak API and MQTT, Project. Design and Development of Realtime IoT Systems.	
UNIT-IV	7 Hours
IoT Problem Identification and IoT PROJECT IMPLEMENTATION & Deployment: IoT Robotics, IoT Drones, IoT in agriculture, IoT in Healthcare, Other areas of IoT. AI based IoT Application Development- Solution framework for IoT Applications, Implementation of Device Integration, Data Acquisition, Organization, Integration and Analytics, Embedding intelligence in IoT application. Natural Language Processing, Image and Video Processing, IoT Product Development & Testing with Project. Mini Project	
Text Books	

1.	Giacomo Veneri , Antonio Capasso , Hands-On Industrial Internet of Things: Create a powerful Industrial IoT infrastructure using Industry 4.0, 29 Nov 2018
2.	O. Hersent, D. Boswarthick, O. Elloumi, "The Internet of Things – Key applications and Protocols", 2nd Edition, Wiley, 2012/ Latest Edition.
3.	Shriram K Vasudevan, "Internet of Things", 2nd Edition, Wiley emerging technology series
Reference Books	
1.	Adrian McEwen and Hakim Cassimally" Designing the Internet of Things, , Wiley Publication, Nov 2013
2.	Simone Cirani," Internet of Things- Architectures, Protocols and Standards", WILEY,2018.
3.	"Hands-On Artificial Intelligence for IoT: Expert machine learning and deep learning techniques for developing smarter IoT systems", by Amita Kapoor. Publisher: Packt Publishing Ltd. 2018.
4.	Rajib Mall, "Real-Time Systems: Theory and Practice," Pearson, 2008.

List of Practical's

1. Introduction to programming language python and its programming development environment
2. Introduction to IoT application development hardware platforms with Raspberry Pi, Jetson Nano and STM 32
3. Sensor interfacing with the target hardware platform and data generation.
4. Actuators interfacing with target platform and controlling the actuators based on sensor data.
5. Control any two actuators connected to the development board using Bluetooth.
6. Read data from sensor and send it to a requesting client. (The client and server should be connected to same local area network.)
7. Introduction to any cloud platform, exploring IoT services.
8. Push sensor data to cloud and data retrieval from cloud
9. Control an actuator through cloud.
10. Access the data pushed from sensor to cloud and apply any data analytics or visualization services.
11. Create a mobile app to control an actuator with any specific application.
12. Identify a problem in your local area or college which can be solved by integrating the things you learned and create a prototype to solve it (Mini Project).
 - a. Design an IoT based air pollution control system which monitors the air pollution by measuring carbon monoxide, ammonia, etc and gives alarm or sends message when the pollution level is more than permitted range.
 - b. Design an IoT based system which measures the physical and chemical properties of the water and displays the measured values.

Annexure-1 (A)

INDIAN KNOWLEDGE SYSTEM	
Course Code: AEC 201 Contact Hours: L-2 T-0 P-0 Course Category: AEC	Credits: 2 Semesters: III/IV

Introduction:

The Indian Knowledge System (IKS) embodies India's deep-rooted scientific, technological, and philosophical heritage. This course offers an interdisciplinary overview of India's contributions to science, engineering, and sustainable development, encouraging critical thinking and innovation rooted in traditional wisdom. The content is aligned with **NEP 2020** to foster an appreciation of Indian culture and its relevance in modern engineering and management education.

Course Objective:

- To introduce students to key contributions of Indian Knowledge System in science, engineering, and technology.
- To explore ancient Indian developments in architecture, metallurgy, water management, mathematics, and mechanics.
- To inspire future innovations by integrating traditional knowledge with modern engineering practices.
- To develop ethical, environmental, and sustainable thinking influenced by Indian philosophy.

Course Outcomes:

Having successfully completed this course, the student will be able to:

CO1: Know about ancient India and its rich glorious past in terms of advancements in engineering and scientific contributions.

CO2: Learn from the ancient literature regarding philosophy, Ayurveda, Engineering and Architecture and their relevance in the modern world.

CO3: Learn Indian knowledge concepts such as Vedic Mathematics, Astronomy, sustainable practices, or mathematical reasoning in design thinking.

CO4: Relate the learnings and value system from Indian Scriptures to the modern life for betterment at personal and professional level

Pedagogy:

- Interactive lectures with illustrations of ancient Indian innovations
- Case studies and experiential learning (documentary reviews etc)
- Reflective writing, short research assignments
- Student presentations and collaborative discussions

Contents

UNIT-I: Introduction to Indian Knowledge Systems		7 Hours
Overview and scope of Indian Knowledge Systems (IKS), IKS in ancient India and in modern India, Introduction to Vedas, A synopsis of the four Vedas, Vedic Corpus, Vedanta, Upanishad, Non-Vedic Philosophical Systems, Wisdom through Ages, Ancient Education System, Ancient Scriptures.		
UNIT-II: Art and Philosophy		7 Hours
Philosophy and Literature (Maharishi Vyas, Manu, Kanad, Parasar, Banabhatta, Nagarjuna and Panini), Understanding Ayurveda, its principles, and its applications in health and well-being, Ethical Engineering from Indian texts (Arthashastra, Manusmriti, Upanishads) Exploring ancient Indian advancements in various fields like architecture and town planning,		
UNIT-III: Sciences and Technologies		9 Hours
Ancient Indian Mathematical Concepts, Number Systems & Measurements, Vedic Mathematics: Patterns and Mental Calculations, Cosmological Numbers, Concepts like kalpas, yugas, manvataras Nyaya (Logic) - Understanding the principles of India logic. Ancient astronomical systems, calendar systems, and celestial phenomena Astronomy and Instruments: Jantar Mantar, Surya Siddhanta; Aryabhata, Bhaskara II Indigenous technologies for construction, measuring, and fabrication		
UNIT-IV Scriptures and Modern Life		7 Hours
Scientific background of Vedic culture, Consciousness, Vedic wisdom, Indian Scriptures like Srimad Bagvad Gita/Ramayana and others and their influence in modern life including Corporate governance, motivation, leadership and others Engineering for welfare: dharma, responsibility, societal benefit, Integrating values into engineering thinking Places and festivals in India and their Scientific importance		
Text Books		
1.	NPTEL course material on “Indian Knowledge Systems”	
2.	IGNCA Publications on Indian Engineering Systems	
Reference Books		
1.	Mysticism A meeting Point of Science and Religion by Dr T D Singh	
2.	Bhagavad Gita As It Is – Srila Prabhupada	
3.	Mahadevan, B., Bhat, Vinayak Rajat, Nagendra Pavana R.N., Introduction to Indian Knowledge System: Concepts and Applications, PHI Publication 2025	

Human Values and Professional Ethics

Course Code	AEC 202	Course Category	AEC
Semester	IV	L-T-P	1-0-2
Teaching Hours/Week	3 Hours	Credits	2

Introduction

This course is a foundation input intended to help students develop right understanding and sensitivity about human values and professional ethics. The focus is on self-exploration and experiential validation of the content (through reflection, dialogue, and application in day-to-day living) rather than only information transfer. The course enables students to examine their aspirations for happiness and prosperity, understand harmony at the levels of self, family, society, nature, and existence, and relate these to ethical professional conduct.

Course Objectives

- Help students appreciate the complementarity between values (right understanding) and skills (professional competence) for sustained happiness and prosperity.
- Enable self-exploration to identify one's aspirations, concerns, and the gap between what is naturally acceptable and what is currently being lived.
- Develop clarity about the 'Self' and the 'Body' and understand the role of feelings, thoughts, and behavior in human conduct.
- Understand harmony in relationships in family and society, and extend the understanding to harmony with nature and sustainable living.
- Strengthen ethical decision-making in professional life through discussion of typical dilemmas, case studies, and holistic approaches.

Course Outcomes (COs)

CO1: Understand the importance of human values and professional ethics in personal, social, and engineering practice.

CO2: Apply self-exploration to achieve harmony in the self and in human relationships.

CO3: Analyze societal and environmental issues from the perspective of sustainability and ethical responsibility.

CO4: Resolve professional ethical dilemmas using value-based and socially responsible reasoning.

Pedagogy: Dialogue-based classroom interaction, guided self-exploration, group discussion, short reflective writing, and case-study analysis.

Unit I	
Introduction to Human Values and Self-Exploration: Need for value education; human aspirations – happiness and prosperity; challenges in present education and society; concept of holistic development; self-exploration as the process for value education; natural acceptance and experiential validation; overview of harmony at the levels of self, family, society, and nature.	8
Unit II	
Harmony in the Self and Human Relationships: Understanding the Self and the Body; happiness and right understanding; feelings, thoughts, and behavior; role of values in personal conduct; harmony in family and society; trust, respect, affection, care, guidance, gratitude, and love; justice and mutual fulfillment in human relationships.	10
Unit III	
Harmony with Society and Nature: Human order and social system; education, health, and justice; interdependence in society; co-existence in nature; natural resources, sustainability, and environmental ethics; responsible consumption and production; role of engineers in ecological balance.	9
Unit IV	
Professional Ethics and Holistic Development: Professional ethics and values; competence, accountability, and integrity; ethical dilemmas in engineering and technology; social responsibility and inclusive development; holistic technologies and sustainable systems; case studies related to engineering ethics, environment, data, safety, and society.	8

Text Books

- R. R. Gaur, R. Sangal, and G. P. Bagaria, A Foundation Course in Human Values and Professional Ethics, 2nd Edition, Excel Books.
- Teachers' Manual for A Foundation Course in Human Values and Professional Ethics (UHV).

Reference Books

- Mike W. Martin and Roland Schinzinger, Ethics in Engineering, McGraw-Hill.
- Charles E. Harris Jr., Michael S. Pritchard, and Michael J. Rabins, Engineering Ethics: Concepts and Cases, Cengage Learning.
- ACM Code of Ethics and Professional Conduct; IEEE Code of Ethics (for classroom discussion).
- AICTE / UHV related resource materials and course modules released from time to time.

Human Values and Professional Ethics

Course Code	AEC 202	Course Category	AEC
Semester	IV	L-T-P	2-0-0
Teaching Hours/Week	3 Hours	Credits	2

Introduction

This course is a foundation input intended to help students develop right understanding and sensitivity about human values and professional ethics. The focus is on self-exploration and experiential validation of the content (through reflection, dialogue, and application in day-to-day living) rather than only information transfer. The course enables students to examine their aspirations for happiness and prosperity, understand harmony at the levels of self, family, society, nature, and existence, and relate these to ethical professional conduct.

Course Objectives

- Help students appreciate the complementarity between values (right understanding) and skills (professional competence) for sustained happiness and prosperity.
- Enable self-exploration to identify one's aspirations, concerns, and the gap between what is naturally acceptable and what is currently being lived.
- Develop clarity about the 'Self' and the 'Body' and understand the role of feelings, thoughts, and behavior in human conduct.
- Understand harmony in relationships in family and society, and extend the understanding to harmony with nature and sustainable living.
- Strengthen ethical decision-making in professional life through discussion of typical dilemmas, case studies, and holistic approaches.

Course Outcomes (COs)

CO1: Understand the importance of human values and professional ethics in personal, social, and engineering practice.

CO2: Apply self-exploration to achieve harmony in the self and in human relationships.

CO3: Analyze societal and environmental issues from the perspective of sustainability and ethical responsibility.

CO4: Resolve professional ethical dilemmas using value-based and socially responsible reasoning.

Pedagogy: Dialogue-based classroom interaction, guided self-exploration, group discussion, short reflective writing, and case-study analysis.

Unit I	
Introduction to Human Values and Self-Exploration: Need for value education; human aspirations – happiness and prosperity; challenges in present education and society; concept of holistic development; self-exploration as the process for value education; natural acceptance and experiential validation; overview of harmony at the levels of self, family, society, and nature.	8
Unit II	
Harmony in the Self and Human Relationships: Understanding the Self and the Body; happiness and right understanding; feelings, thoughts, and behavior; role of values in personal conduct; harmony in family and society; trust, respect, affection, care, guidance, gratitude, and love; justice and mutual fulfillment in human relationships.	10
Unit III	
Harmony with Society and Nature: Human order and social system; education, health, and justice; interdependence in society; co-existence in nature; natural resources, sustainability, and environmental ethics; responsible consumption and production; role of engineers in ecological balance.	9
Unit IV	
Professional Ethics and Holistic Development: Professional ethics and values; competence, accountability, and integrity; ethical dilemmas in engineering and technology; social responsibility and inclusive development; holistic technologies and sustainable systems; case studies related to engineering ethics, environment, data, safety, and society.	8

Text Books

- R. R. Gaur, R. Sangal, and G. P. Bagaria, A Foundation Course in Human Values and Professional Ethics, 2nd Edition, Excel Books.
- Teachers' Manual for A Foundation Course in Human Values and Professional Ethics (UHV).

Reference Books

- Mike W. Martin and Roland Schinzinger, Ethics in Engineering, McGraw-Hill.
- Charles E. Harris Jr., Michael S. Pritchard, and Michael J. Rabins, Engineering Ethics: Concepts and Cases, Cengage Learning.
- ACM Code of Ethics and Professional Conduct; IEEE Code of Ethics (for classroom discussion).
- AICTE / UHV related resource materials and course modules released from time to time.